

Bangladesh University of Engineering and Technology

Department of Computer Science and Engineering

L-3/T-2

July 2025

CSE 311: Data Communication

Time: 25 minutes**Marks: 20**

Student Name: _____

Student ID: _____

1. Consider a message signal $g(t)$ whose Fourier Transform is $G(f)$. (10)
 - (a) If the phase of $g(t)$ is shifted by $2\pi f_0 t$ in time domain, what will happen in frequency domain? Briefly explain using Fourier Transform. (4)
 - (b) Suppose $g(t)$ is modulated by $\cos 2\pi f_0 t$. Then the phase of modulated signal is further changed by θ_0 . Find the Fourier Transform of the new signal. (3)
You may use the result of the Fourier Transform from question 1(a).
 - (c) In question 1(b), we consider $g(t) = 1$ and $\theta_0 = 0$. Find the Fourier Transform of this signal. Also, sketch the graph of the Fourier Transform. (3)
Hint: $1 \iff \delta(f)$

2. Consider a signal $g(t)$ whose spectrum is band-limited to 5 kHz . It is sampled according to Sampling Theorem. (10)
- (a) Find the Nyquist Rate and Nyquist Interval for this signal. (5)
- (b) If we use 7 kHz as the sampling frequency for this signal, find whether it can be perfectly recovered or not using a Low Pass Filter. Briefly explain with necessary figures. (5)